

Comparative Analysis of Phase Frequency Detector for Phase-Locked Loops

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Abstract

This paper presents phase frequency detector using different gates like AND, NOR and NAND and comparative simulation result analysis is carried out. The simulation is carried out using 0.18 μm CMOS process technology with supply voltage 1.8 V. NAND based PFD has lowest power consumption then other and it is 4.2587 μW at 1 GHz Frequency and dead zone to 18 ps which means having approximately zero dead zone problem. So, it is suitable for low power application.

Keywords: Dead zone, glitch period, phase-locked loops, phase frequency detector

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INTRODUCTION

A PLL is a feedback system that compares the output phase with the input phase. The comparison is performed by a “Phase Detector” (PD) [1]. A PLL is a circuit that causes a particular system to track with another one. More precisely, a PLL is a circuit synchronizing an output signal which are generated by an oscillator with a reference or input signal in frequency as well as in phase [2].

The phase-locked loop (PLL) plays the role of generating a clock signal that is usually a multiple of a reference clock and synchronized with the reference clock in phase. In most cases, the charge pump PLL (CP-PLL) is used due to its high frequency range and simple structure. In the PLL, the phase frequency detector (PFD) compares the rising edges of the reference clock and the voltage-controlled oscillator (VCO) clock, and generates a lead signal when the reference phase is leading or a lag signal when

the reference phase is lagging. The phase difference detected in the PFD passes through the loop filter to control the VCO. As the phase difference critically affects the overall characteristics of the PLL such as lock-in time and jitter performance, the PFD should be designed to work accurately for any phase difference [3].

A PLL is feedback system that fixes relation between out-put clock phase and input clock phase. Actually, phase of both input signal and output signal are synchronized or locked, hence name called “Phase Locked Loop” as shown in Figure 1 [4].

Figure 1 consists of four blocks i.e. phase detector, loop filter, voltage controlled oscillator and frequency divider. This is called 1st order PLL or Type 1 PLL. It suffers from limited acquisition range. So, to eliminate this problem, Phase frequency detector along with charge pump is re-placed by phase detector.

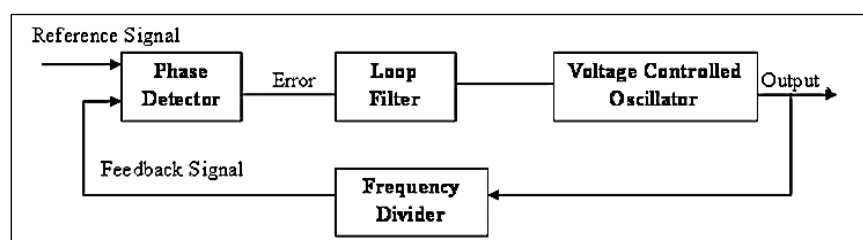


Fig. 1: PLL Block Diagram [4].

CIRCUIT ARCHITECTURE

PFD using AND Gate

The output of the PFD depends on both the phase and frequency of the inputs. This type of the phase detector is also termed as the sequential detector. Phase frequency detector (PFD) is a digital circuit detecting phase or frequency difference between reference clock and voltage controlled oscillator (VCO) clock/feedback signal and generates output signal if frequency of VCO is to be increased or decreased. PFD drives the charge pump (CP) and adjust the amount of the current to be injected to or from the loop filter capacitor (Figure 2). If there is a phase difference between the two signals, it generates up or down synchronized signals to the charge pump.

The operation of this circuit is based on two D-type flip-flop and simple AND gate. Each flip-flop has the D-input wired high. Under this condition, the flip-flop with a low Q output will transition to high on the next rising edge of its clock input. Also, if such a transition occurs when Q is high, then there will no change in the flip-flop state. A high signal on a reset input will force Q low as soon as the reset signal is applied. Finally, a logically high on both of the Q output causes the resetting of both the flip-flops. The PFD generates two outputs that are not complementary. The output signal depends not only on the phase error, but also on the

frequency error. If the frequency, ω_A , of the input A is less than that of input B, ω_B , then the PFD produces positive pulses at QA, while QB remains zero.

Conversely, if $\omega_A > \omega_B$, then positive pulses appear at QB while QA=0. If $\omega_A = \omega_B$, then circuit generates pulses at either QA or QB with a width equal to the phase difference between the two inputs. Thus, the average value of the QA - QB is proportional to the frequency or the phase difference between the inputs at data and dclock. The outputs QA and QB are usually called the up and down signals. The UP pulse is the difference between the phases of the two clock signals. This UP pulse indicates that the feedback signal needs to speed up with the reference signal. In the second case, when the feedback signal is leading the reference clock signal, the DOWN pulse represents the difference between the phases of the two clock signals. A circuit design of PFD using AND gate which consisting of only 22 transistors is shown in Figure 3.

Simulation Results in 0.18 μm Technology

Simulation results of AND gate based PFD at 20 ns in lock condition is shown in Figure 4. Figures 5 and 6 shows the waveforms at clock leading and lagging condition respectively at 20 ns. Dead zone of this circuit is 62.225 ps at 50 MHz frequency which is shown in Figure 7.

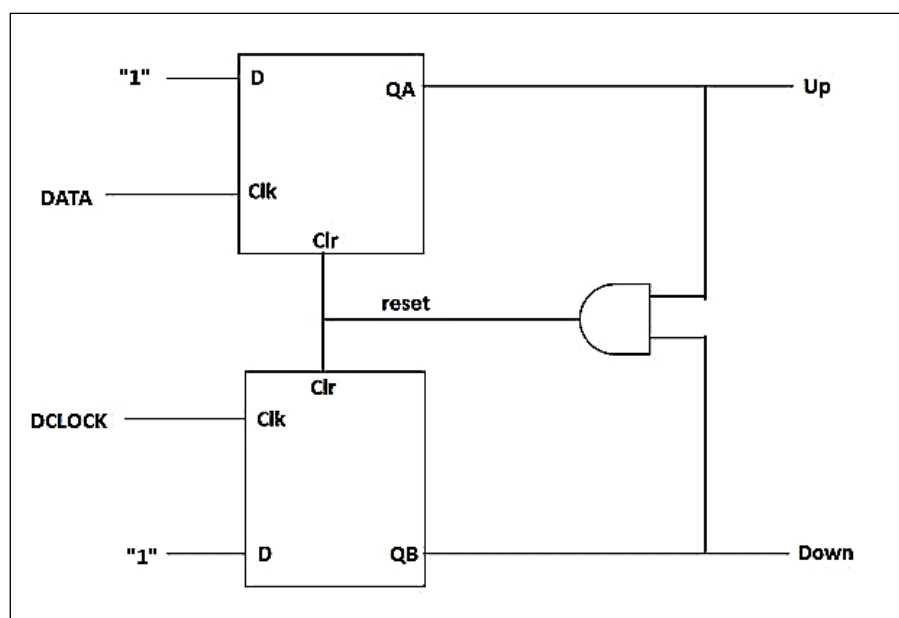


Fig. 2: A Basic Block Diagram of AND Gate Based Phase Frequency Detector [5].

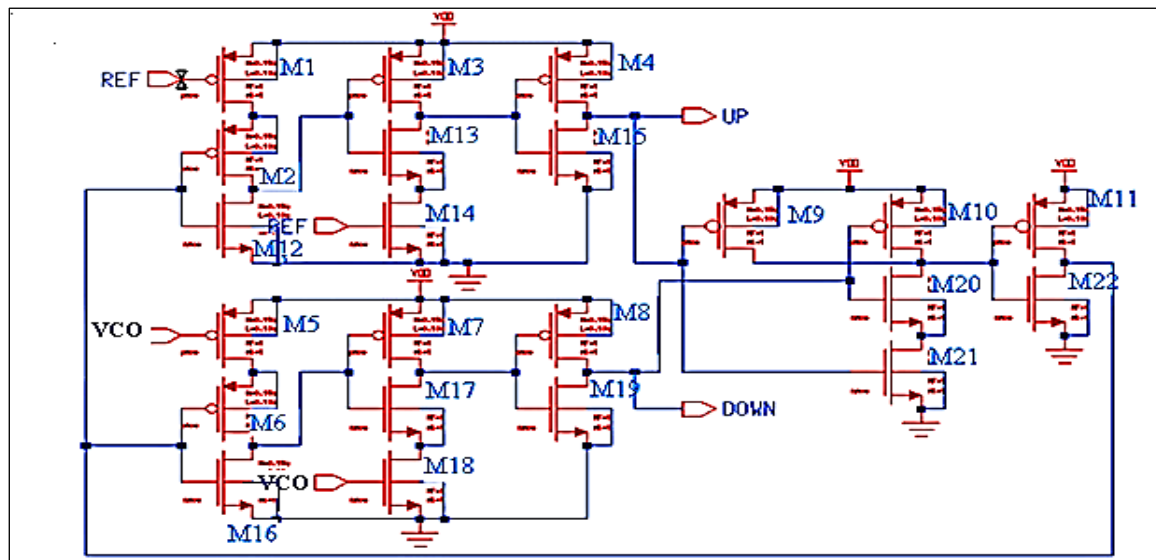


Fig. 3: AND Gate Based PFD Schematic [5].

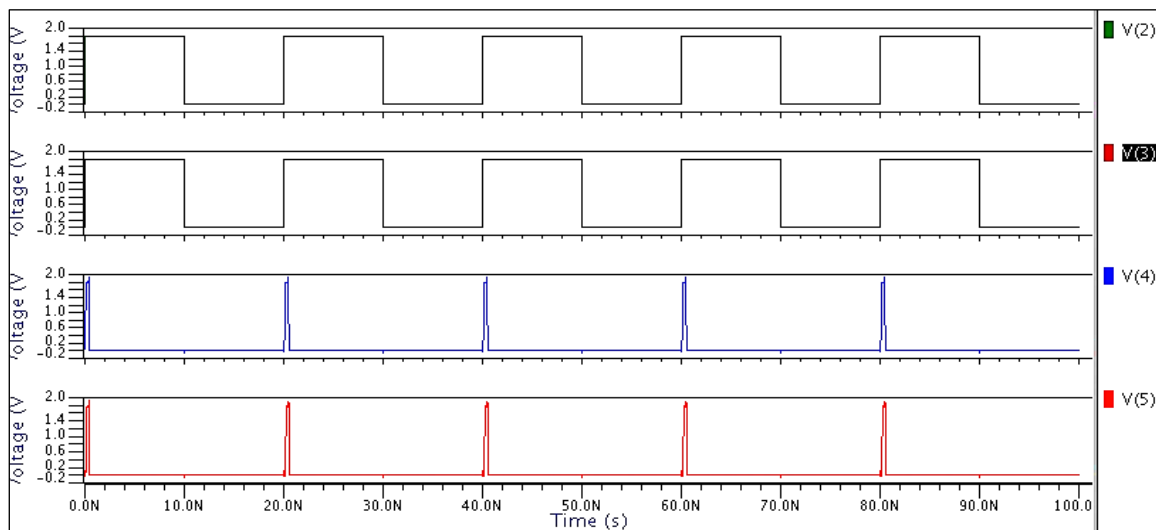


Fig. 4: Basic PFD using AND GATE at 20 ns (Lock Condition).

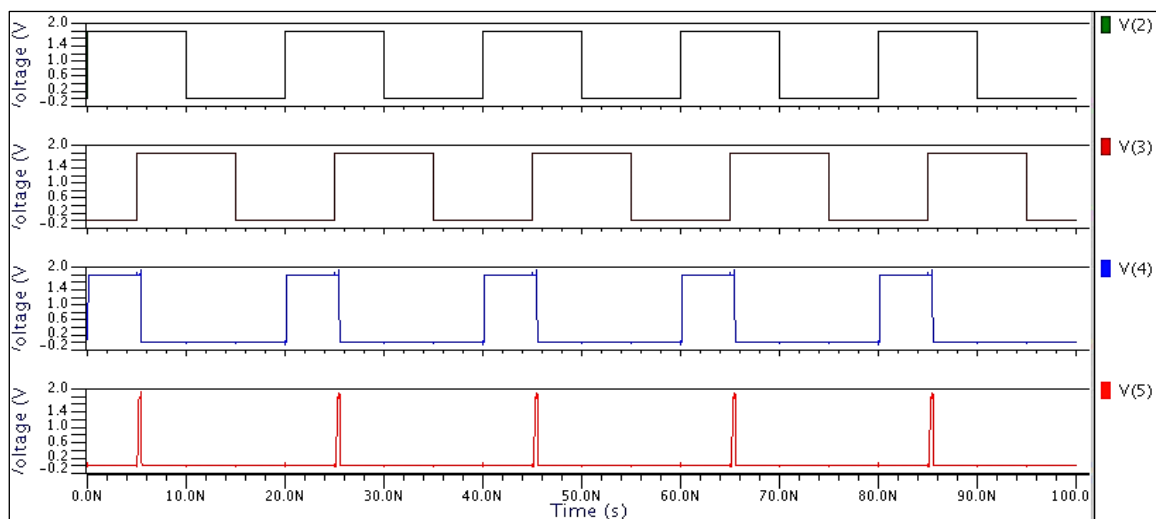


Fig. 5: PFD using AND GATE at 20 ns (CLK Leading Condition).

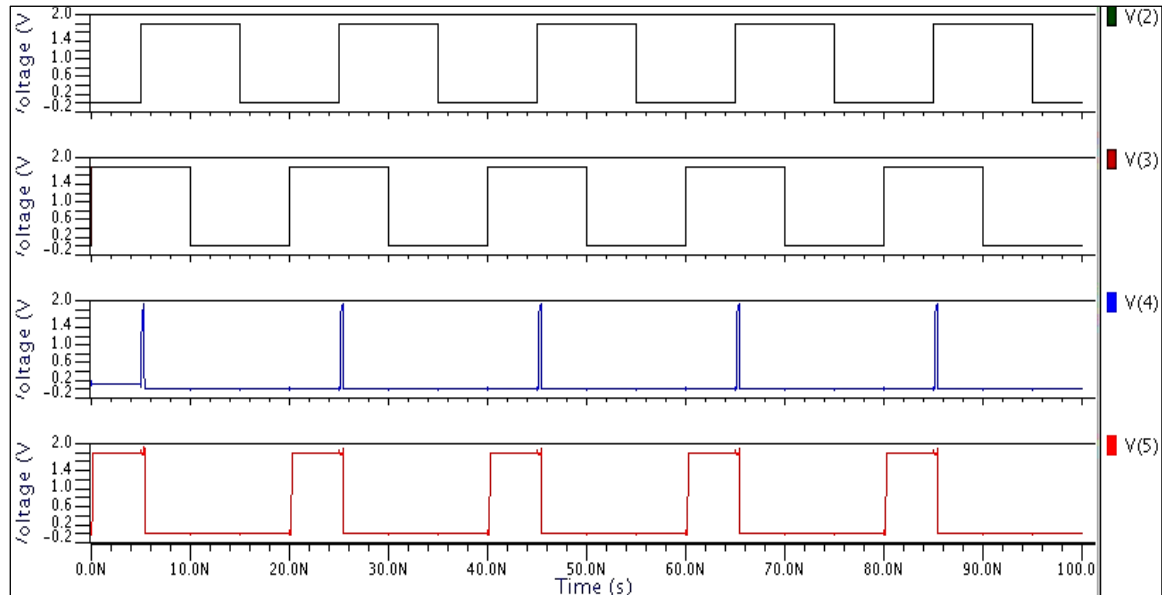


Fig. 6: Basic PFD using AND GATE at 20 ns (CLK Lagging Condition).

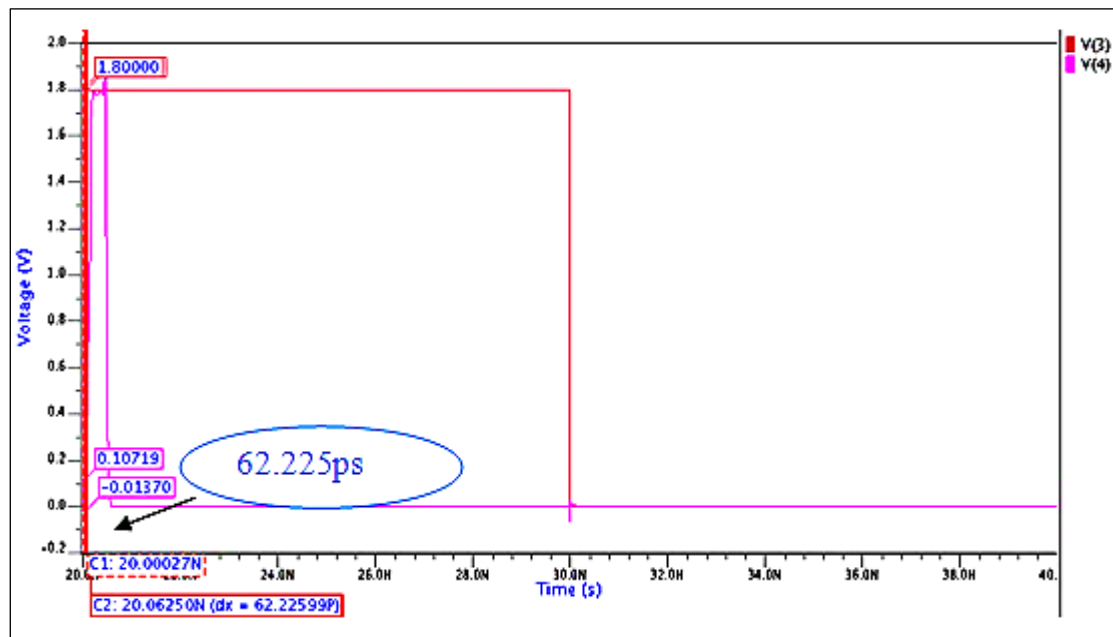


Fig. 7: Dead Zone of AND GATE PFD at 50 MHz Frequency.

Comparative Analysis of AND Gate based PFD

Table 1 shows the performance comparison of PFD using AND gate. It is seen that power consumption at 1 GHz is found to be 7.7899 μ W. Dead zone is reduced to 45 ps. Also Glitch period is reduced to 321ps. Table 2 shows the performance comparison of PFD using AND gate at 0.18 μ m cmos process with 1.8 V supply voltage at different frequencies. It is seen that if frequency is increased then power consumption is also increase but dead zone and glitch period both are reduced.

PFD using NOR Gate

In this type of PFD, the reset path is modified either using nor gate. The inputs signals–REF and VCDL are directly connected to the input signal of D flip flop as shown in the Figure 8. The NOR gate in the reset path so it contains less transistor as compared to and gate in the reset path so the dead zone is less. With the modified circuit of D flip flop the reset time is reduced and so the jitter is reduced. The circuit of D flip flop is modified. A circuit design of PFD using NOR gate which consisting of only 20 transistors is shown in Figure 9.

Table 1: Design Specification of AND Gate PFD.

Parameter	Specification [5]	Design Results
Technology	0.18 μ m	0.18 μ m
Input frequency	1 GHz	1 GHz
Transistor count	22	22
Power consumption	0.29125 μ W	7.7899 μ W
Dead zone	50 ps	45 ps
Glitch Period	660 ps	321 ps

Table 2: Simulation Result in 0.18 μ m Technology.

Parameter	AND Based PFD	AND Based PFD	AND Based PFD
Technology	0.18 μ m	0.18 μ m	0.18 μ m
Input Frequency	50MHz	500MHz	1GHz
Transistor count	22	22	22
Power Consumption	3.7672 μ W	5.2995 μ W	7.7899 μ W
Dead zone	62ps	50ps	45ps
Glitch Period	468ps	359ps	321ps

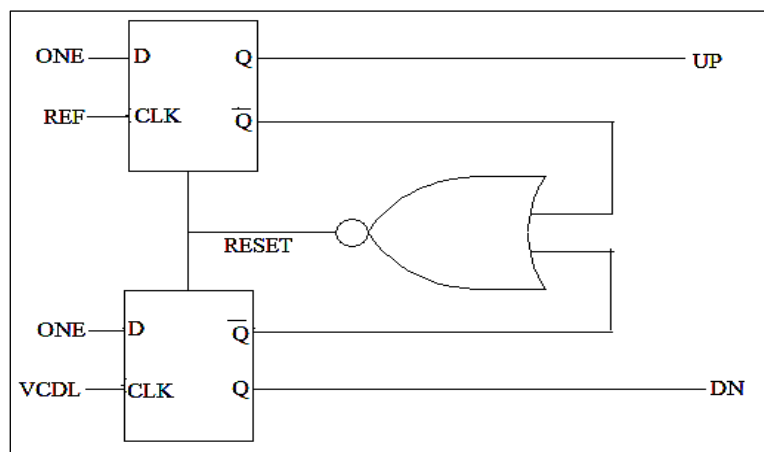


Fig. 8: A Basic Block Diagram of NOR Gate Based Phase Frequency Detector [5].

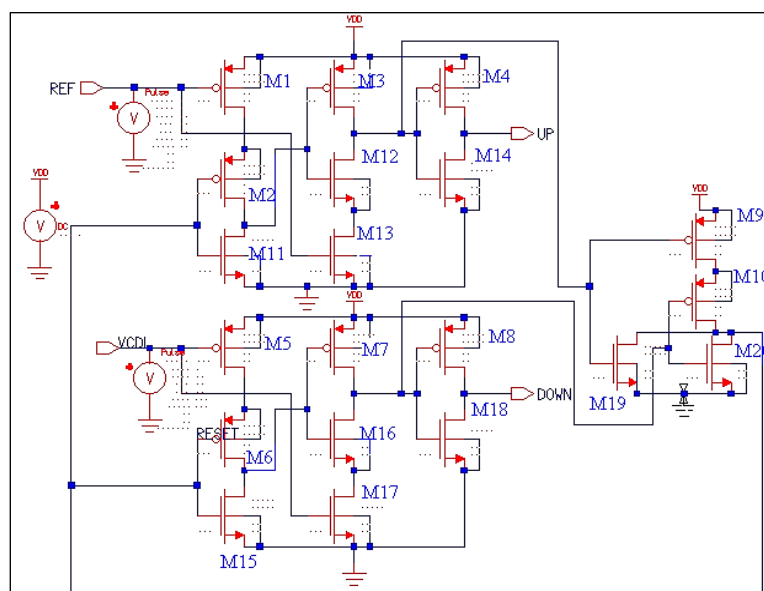


Fig. 9: NOR Gate based PFD Schematic [5].

Simulation Results in 0.18 μm Technology

Simulation results of NOR gate based PFD at 20 ns in Lock Condition is shown in Figure 10. Figures 11 and 12 shows the waveforms at clock leading and lagging condition respectively at 20 ns. Dead zone of this circuit is 54.234 ps at 50 MHz frequency, which is shown in Figure 13.

Comparative Analysis of NOR Gate Based PFD

Table 3 shows the performance comparison of

PFD using NOR gate. It is seen that power consumption at 1 GHz is found to be 6.1316 μW . Dead zone is reduced to 39 ps. Also Glitch period is reduced to 292 ps.

Table 4 shows the performance comparison of PFD using NOR gate at 0.18 μm cmos process with 1.8 V supply voltage at different frequencies. It is seen that if frequency is increased then power consumption is also increase but dead zone and glitch period both are reduced then PFD using AND gate.

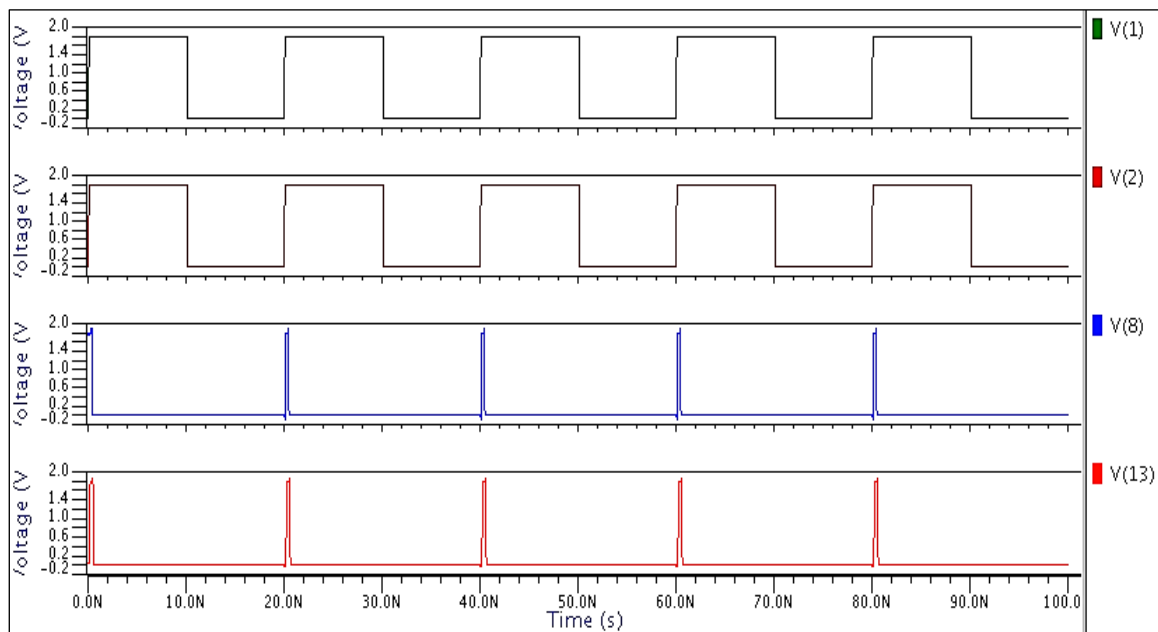


Fig. 10: PFD using NOR GATE at 20 ns (Lock Condition).

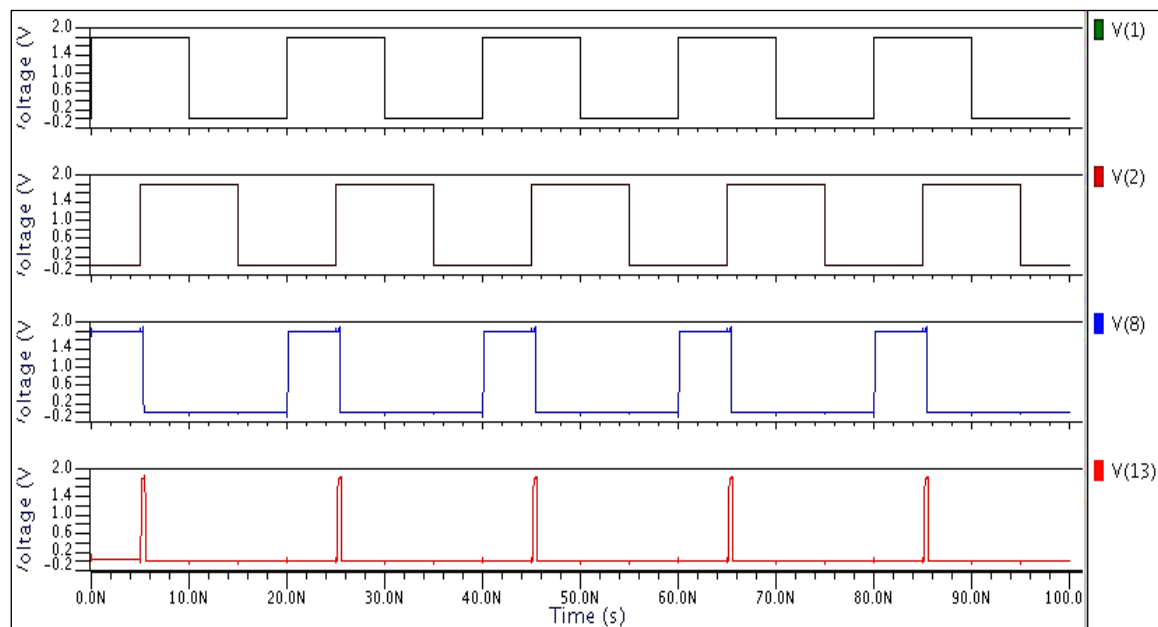


Fig. 11: PFD using NOR GATE at 20 ns (CLK Leading Condition).

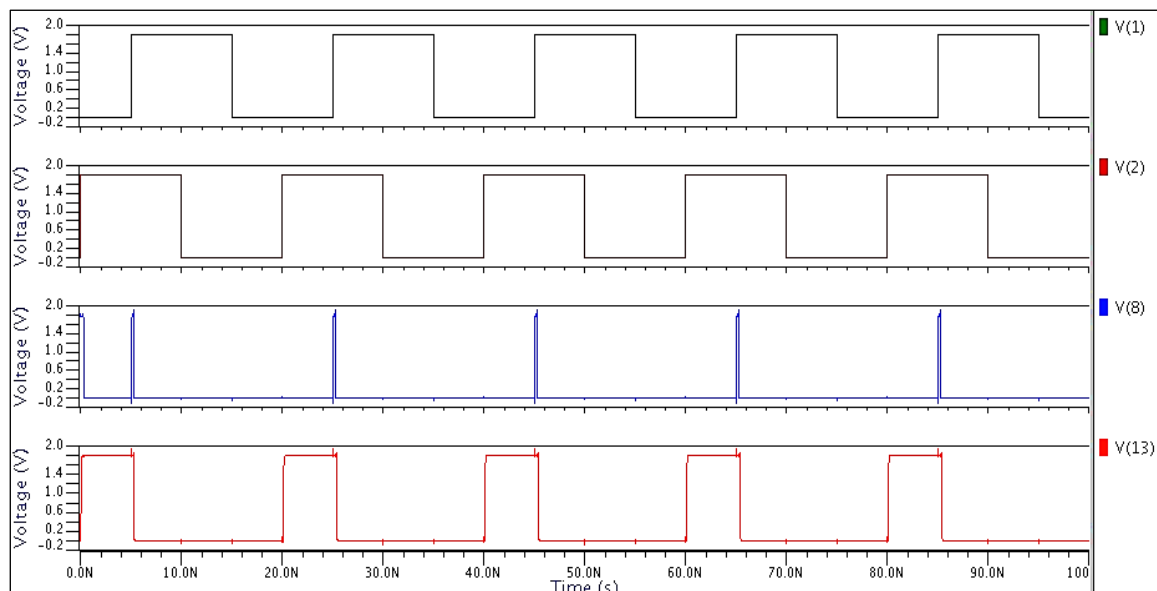


Fig. 12: PFD using NOR GATE at 20 ns (CLK Lagging Condition).

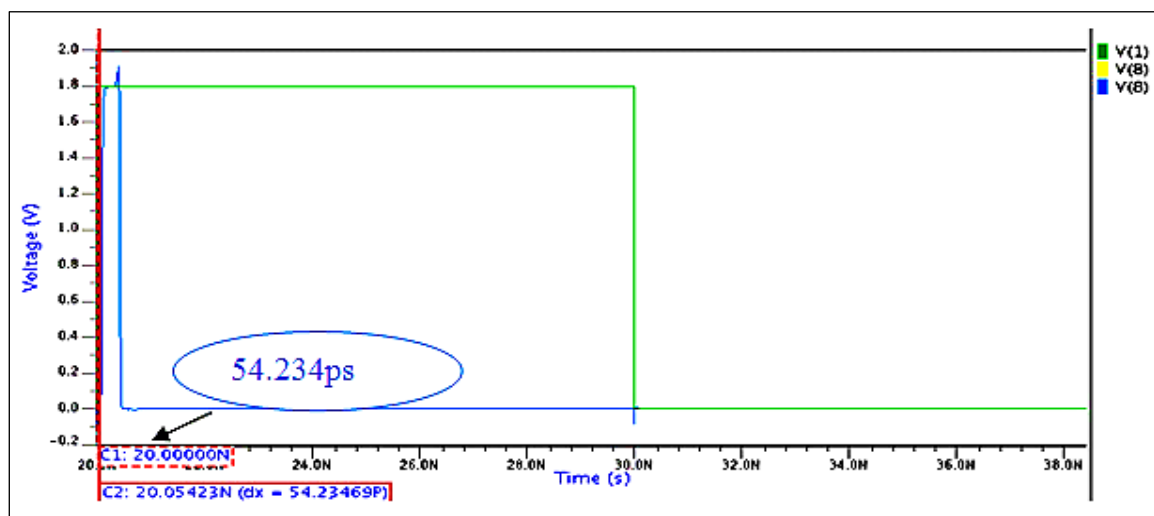


Fig. 13: Dead Zone of PFD using NOR GATE at 50 MHz Frequency.

Table 3: Design Specification of NOR Gate PFD.

Parameter	Specification [5]	Design Results
Technology	0.18 μm	0.18 μm
Input frequency	1 GHz	1 GHz
Transistor count	20	20
Power consumption	0.25719 μW	6.1316 μW
Dead zone	40 ps	39 ps
Glitch Period	540 ps	292 ps

Table 4: Simulation Result in 0.18 μm Technology.

Parameter	NOR Based PFD	NOR Based PFD	NOR Based PFD
Technology	0.18 μm	0.18 μm	0.18 μm
Input Frequency	50 MHz	500 MHz	1 GHz
Transistor count	20	20	20
Power Consumption	3.0715 μW	5.2995 μW	6.1316 μW
Dead zone	54 ps	47 ps	39 ps
Glitch Period	375 ps	321 ps	292 ps

PFD Using NAND Gate

The basic Phase frequency detector is composed by two D-FF and an NAND logic gate (Figure 14). In the NAND based PFD, circuit consists of two resettable, edge triggered D flip flops with their D inputs tied to logic 1 and a NAND gate in the reset path. The explanation of the general operation of the PFD begins by describing the initial state of the device. First, the UP and DN signals are reset to low or zero and assume both the data frequency signal and the dclock signal are high or one. Additionally, the data frequency waveform is slightly leading the dclock waveform. When a falling edge occurs on the data input, the high or one on the D input is transmitted to the Q output or UP. A short time later, the dclock waveform experiences a falling edge and the Q output or DN of the other flip flop is set. Once both UP and DN are high or ONE, the NAND gate experiences a transition to force the Reset signal to zero. The flip flops are designed so that zeros on the Reset signal resets the Q outputs to zero. A circuit design of PFD using NOR gate which consisting of only 16 transistors is shown in Figure 15.

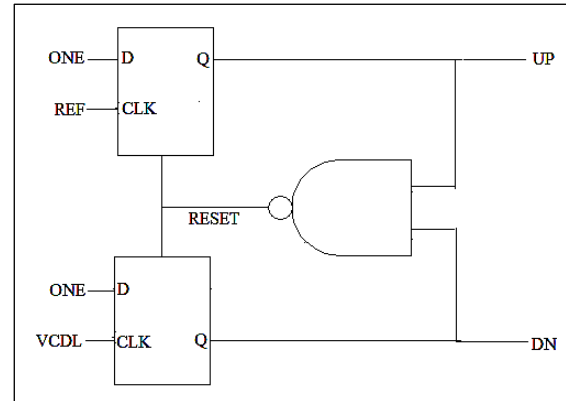


Fig. 14: A Basic Block Diagram of NAND Gate based Phase Frequency Detector [5].

The PFD is implemented with True Single Phase Clocked logic as shown in Figure 15. The design strategy is to minimize the number of transistors and the amount of power consumed. However, not all of the transistors can be implemented with minimum width such as those involved in the reset operation. In Table 5, the transistor sizes for the PFD are given. Note that since the operating frequency of the PFD is quite low compared to the cutoff frequency of the technology process, most transistors can be made minimum size.

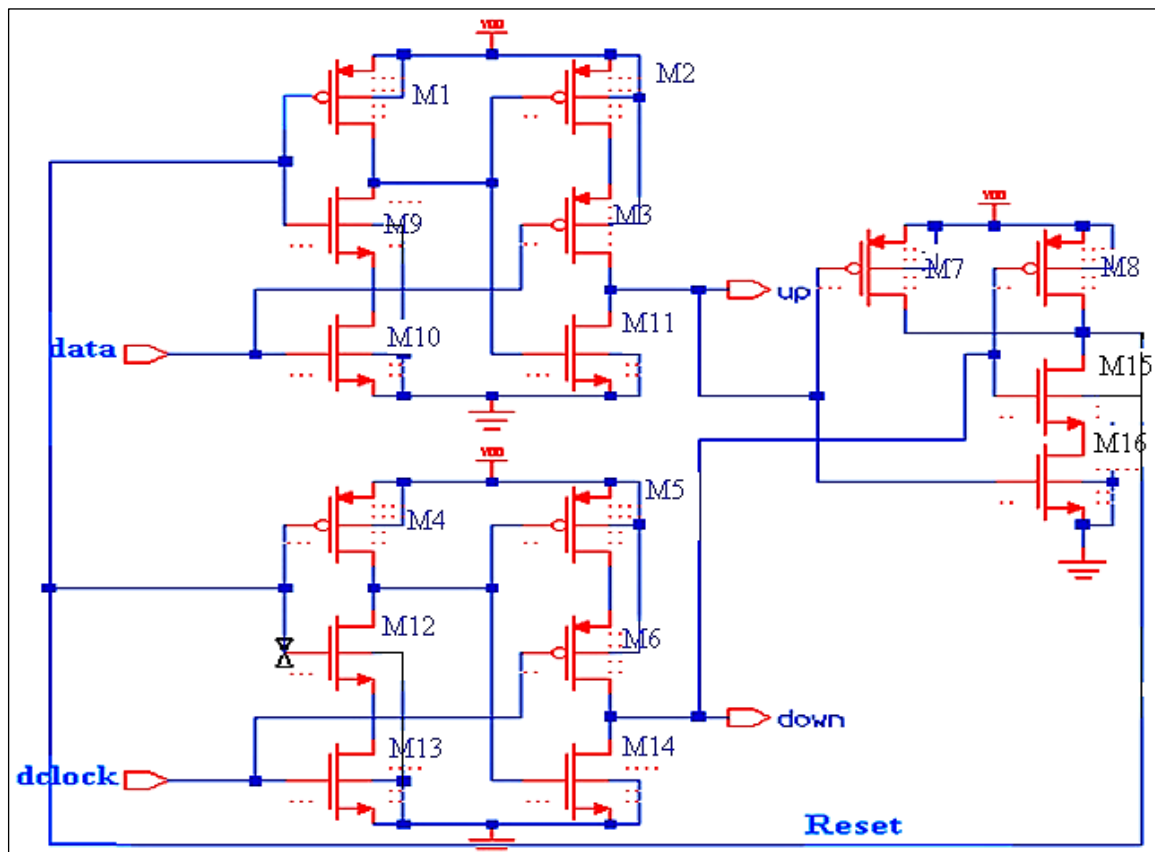


Fig. 15: PFD using NAND GATE Schematic [5].

Table 5: Transistor size for NAND Gate PFD.

Transistor	W/L Ratio
M9	0.27u/.18 u
M10	0.27u/.18 u
M11	2.2u/.18 u
M12	0.27u/.18 u
M13	0.27u/.18 u
M14	2.2u/.18 u
M15	1u/.18 u
M16	1u/.18 u

The width of PMOS transistor is 3 times higher than the NMOS.

Simulation Results in 0.18 μ m Technology

Simulation results of NAND gate based PFD at 20 ns in lock condition is shown in Figure 16. Figures 17 and 18 shows the waveforms at clock leading and lagging condition respectively at 20 ns. Dead zone of this circuit is 22.22 ps at 50 MHz frequency which is shown in Figure 19.

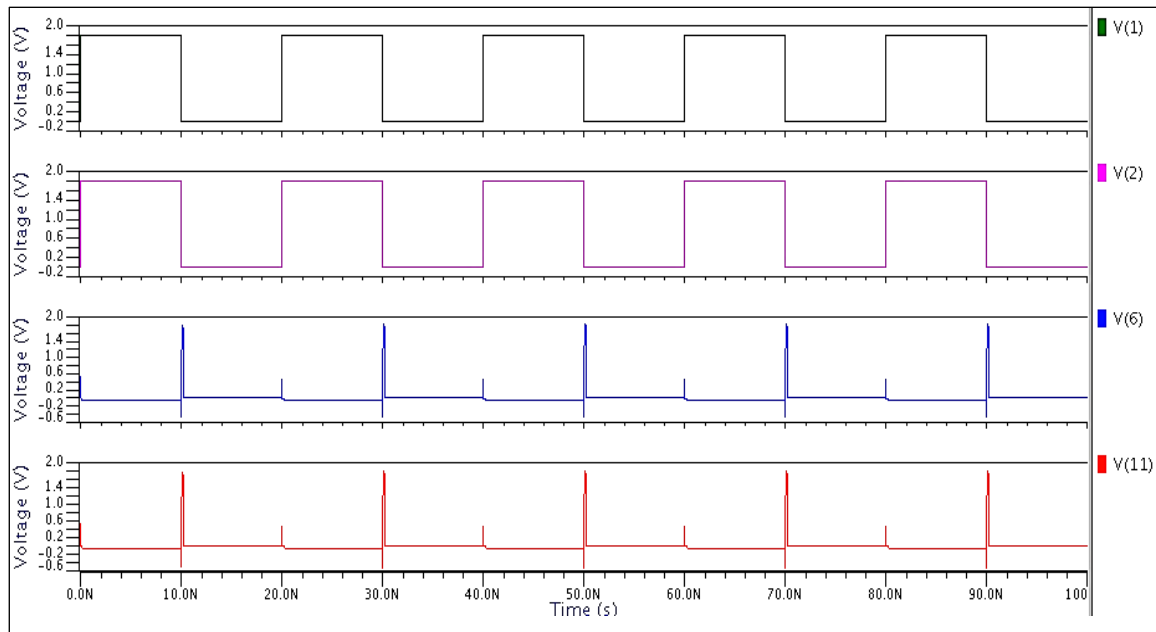


Fig. 16: PFD using NAND GATE at 20 ns (Lock Condition).

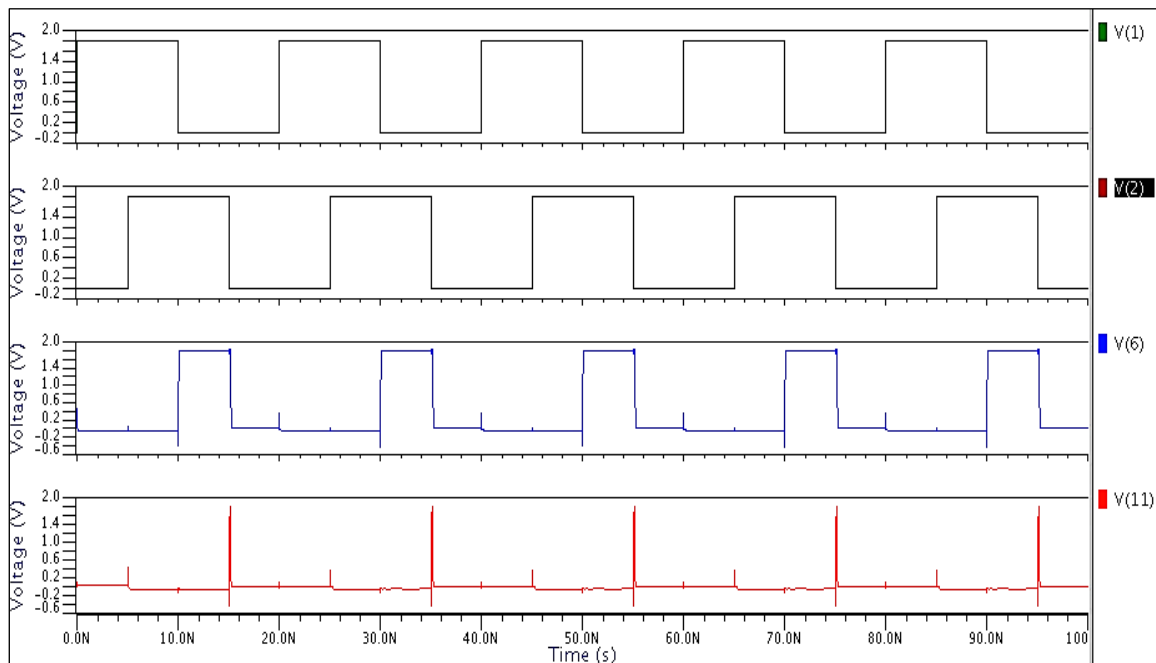


Fig. 17: PFD using NAND GATE at 20 ns (CLK Leading Condition).

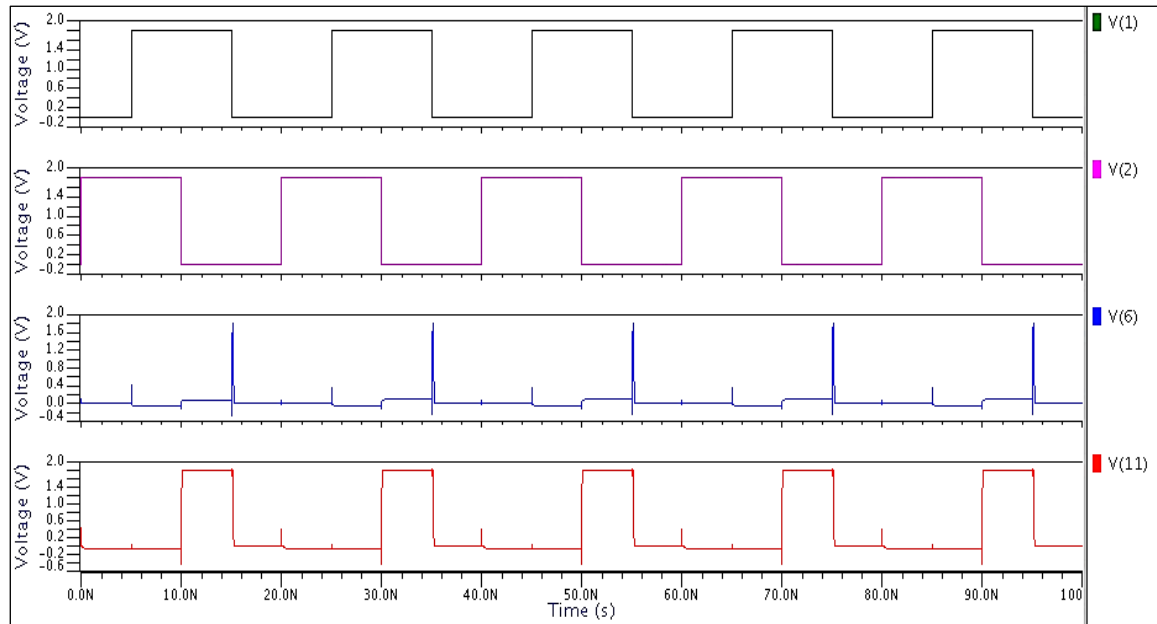


Fig. 18: PFD using NAND GATE at 20 ns (CLK Lagging Condition).

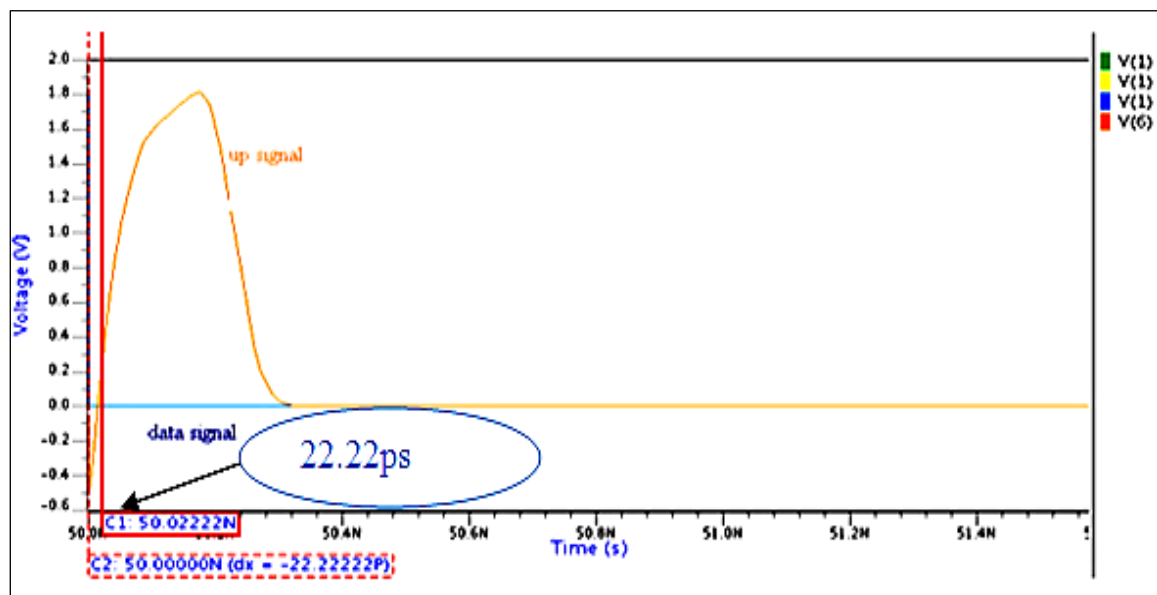


Fig. 19: Dead Zone of PFD using NAND GATE at 50 MHz Frequency

Comparative Analysis of NAND Gate based PFD

Table 6 shows the performance comparison of PFD using NAND gate. It is seen that power consumption at 1 GHz is found to be 4.2587 μ W. Dead zone is reduced to 18 ps. Also Glitch period is reduced to 245 ps.

Table 7 shows the performance comparison of PFD using NAND gate at 0.18 μ m cmos process with 1.8 V supply voltage at different frequencies. It is seen that if frequency is increased then power consumption is also increase but dead zone and glitch period both

are reduced then PFD using AND gate and NOR gate.

Comparisons of AND, NOR, NAND base PFD

Table 8 depicts various parameters for the different PFDs at 1 GHz frequency. It is seen that NAND based PFD has minimum power consumption at same frequency applied to all other PFDs. Also it has low dead zone and glitch period. Also, it has less number of transistors that's why it occupy less area compared to other two PFDs.

Table 6: Design Specification of NAND Gate PFD.

Parameter	Specification [5]	Design Results
Technology	0.18 μm	0.18 μm
Input frequency	1 GHz	1 GHz
Transistor count	16	16
Power consumption	0.21577 μW	4.2587 μW
Dead zone	25 ps	18 ps
Glitch Period	380 ps	245 ps

Table 7: Simulation Result in 0.18 μm Technology.

Parameter	NAND Based PFD	NAND Based PFD	NAND Based PFD
Technology	0.18 μm	0.18 μm	0.18 μm
Input Frequency	40 MHz	500 MHz	1 GHz
Transistor count	16	16	16
Power Consumption	2.954 μW	3.4929 μW	4.2587 μW
Dead zone	22 ps	20 ps	18 ps
Glitch Period	286 ps	260 ps	245 ps

Table 8: Comparisons of Three PFD at 1 GHz Frequency in 0.18 μm Technology.

Parameter	AND Based PFD	NOR Based PFD	NAND Based PFD
Technology	0.18 μm	0.18 μm	0.18 μm
Input Frequency	1 GHz	1 GHz	1 GHz
Transistor count	22	20	16
Power Consumption	7.7899 μW	6.1316 μW	4.2587 μW
Dead zone	45 ps	39 ps	18 ps
Glitch Period	321 ps	292 ps	245 ps

CONCLUSIONS

Minimizing of power consumption is essential performance VLSI Design. This paper compares the different parameters of PFD by different logic gates. We can see from the simulation of PFD using AND gate it consumes more power than all other PFDs and also more dead zone and glitch period. NOR gate based PFD also consume more power than NAND based PFD. So for low power consumption, dead zone and glitch period we use NAND gate based PFD.

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